What is RAG?

Retrieval-Augmented Generation — a sourced one-page primer

Source 1 — Definition

Retrieval-Augmented Generation (RAG) is a technique that improves the answers of a large language model by giving it relevant information retrieved from an external knowledge source at the moment a question is asked. Instead of relying only on what the model learned during training, a RAG system first searches a document collection and then feeds the most relevant passages to the model as extra context.

Source 2 — Why RAG matters

A plain language model can only answer from its fixed training data, which may be outdated and does not include your private documents. RAG lets the same model answer questions about your own files without retraining it. This reduces hallucination, keeps answers current, and allows the system to cite the sources it used.

Source 3 — The five steps of how RAG works

RAG works in five steps. First, ingest: documents are split into small chunks of text. Second, embed: each chunk is converted into a numeric vector by an embedding model. Third, store: those vectors are saved in a vector database. Fourth, retrieve: the user's question is embedded and used to find the most similar chunks. Fifth, generate: the retrieved chunks are passed to the language model, which composes a grounded answer.

Source 4 — Key components

A RAG system has four key components. The embedding model turns text into vectors, for example mxbai-embed-large. The vector database stores vectors and finds similar ones quickly, for example Qdrant. The language model generates the final answer from the retrieved context, for example Llama 3.2. The orchestrator connects these pieces into a pipeline, for example an n8n workflow.

Source 5 — A simple analogy

Think of an open-book exam. A student who memorised everything, like a plain language model, may misremember details. A student allowed to look up the exact relevant page before answering, like a RAG system, gives more accurate and verifiable answers, even for material never seen before.

Source 6 — Common uses

RAG is widely used for question answering over private documents, customer support assistants grounded in a company knowledge base, research tools that answer from a library of papers, and internal help desks that answer from policies and manuals. In each case the model stays the same and only the retrieved documents change.

Test file for a Retrieval-Augmented Generation pipeline. Each paragraph above is a distinct labelled source. After ingesting this file, ask the chatbot questions such as “What are the five steps of how RAG works?” (should draw on Source 3) or “What is the open-book exam analogy?” (should draw on Source 5), and check whether the retrieved answer matches the right source.